MCO-1D Metachronal Cascade Ornithopter

Wing Root Stress Test — Rattan Rod Simulation Table

Setup: clamp 28" × 1" rattan rod horizontally. Hang listed weight at stated distance from clamp. The rod represents the wing spar; the clamp is the wing root.

Flight Condition	Load factor	Flap Hz	Root moment (N·cm)	Weight at 7" inertia tests	Weight at 10" combined tests	Weight at 12" aero tests	Root stress (MPa)
1g level cruise	1.0g	—	144	394 g	276 g	230 g	0.896
2g banked turn 45°	2.0g	—	288	1221 g	855 g	712 g	1.793
3g steep bank 70°	3.0g	—	433	2048 g	1433 g	1195 g	2.689
3.5g crab / hard manoeuvre	3.5g	—	505	2461 g	1723 g	1436 g	3.137
3.8g utility limit (static)	3.8g	—	548	2709 g	1896 g	1580 g	3.406
4g stoop pullout (static)	4.0g	—	577	2874 g	2012 g	1677 g	3.585
Flap inertia only 1 Hz	—	1	33	rod only	rod only	rod only	0.207
Flap inertia only 2 Hz	—	2	133	332 g	232 g	193 g	0.828
Flap inertia only 3 Hz	—	3	300	1287 g	901 g	750 g	1.864
1g cruise + 1 Hz flap	1.0g	1	178	585 g	**410 g**	341 g	1.103
1g cruise + 2 Hz flap	1.0g	2	277	1158 g	**811 g**	676 g	1.725
1g cruise + 3 Hz flap	1.0g	3	444	2113 g	**1479 g**	1233 g	2.760
3.8g limit + 3 Hz flap	3.8g	3	848	4428 g	**3100 g**	2583 g	5.270
4g stoop + 3 Hz — WORST CASE	4.0g	3	**877**	4593 g	**3215 g**	2679 g	5.449

Key: Aerodynamic only | Inertial / flap only | Combined (use 10" column) | Worst case

1. Weights are **additional** to the rod's own 216 g. The rod hanging horizontally already contributes 75 N·cm at the root.

2. 7" column: pure flap inertia tests (oscillate at stated Hz). **12" column**: static lift (aerodynamic) tests. **10" column**: single-weight combined tests.

3. Rattan flexural strength ~80–120 MPa. Max stress 5.45 MPa — safety factor ~15x. Rod will not break.

4. W1 (772 mm) is modelled — most loaded wing. W2–W4 require 2–8% less weight for the same load case.